

Doc. 269 – FFGFT and RA/PMT:
a bidirectional structural comparison

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Contents

.1	The RA/PMT scheme in brief	1
.2	Direction A: FFGFT evaluates RA/PMT (six-axis operator)	2
.3	Direction B: the RA/PMT grid applied to FFGFT	3
.4	Synthesis: two opposite poles of the same question	4

Purpose

This document compares FFGFT (Fundamental Fractal Geometric Field Theory, T^4 time-mass duality) with the RA/PMT scheme of J. T. Guevara (*Regional Arenas / Perception Mechanism Theory*, 2026, three parts: consciousness, fundamental physics, cosmology).

The comparison uses the same explicit comparison operator as Doc. 265 – but applied here *in both directions*:

- **Direction A:** FFGFT evaluates RA/PMT using the six-axis operator (Doc. 265).
- **Direction B:** the RA/PMT grid itself ($RA4 \rightarrow RA1.5 \rightarrow RA2 \rightarrow RA1$) is applied to FFGFT – just as Guevara applies it to QFT, GR and emergent spacetime.

Applying it both ways is exactly the point of an exposed operator (Doc. 265): each side may render the other with its own tool, and both renderings remain examinable side by side.

Up front – the most important caveat: FFGFT and RA/PMT are *not on the same level*. FFGFT is a physical theory with one free parameter ($\xi = 1/7500$) from which concrete numbers follow (CMB peak ratios, H_0 , mass relations). RA/PMT is, in Guevara's own words, "*not a new physical theory*" – a meta-scheme for classifying theories. The comparison is therefore not between two competing models of the same system (as Doc. 265 was for FFGFT/Blume-Maleev), but between a theory and a theory-ordering scheme. This shapes every finding below.

.1 The RA/PMT scheme in brief

RA/PMT decomposes every theory into five layers:

Layer	Function
RA4	Blueprint – the structured space of all <i>possible</i> configurations (prior to any realization).
RA3	Residual – the theory’s undefined terms, explicitly declared as unresolved rather than hidden.
RA2	Mapping – the rules/equations that unfold a chosen blueprint into physical appearance.
RA1.5	Mechanism – the <i>selector</i> that picks one configuration from RA4. (HPM for consciousness, PSM for physics, CSM for cosmology – the same role, per Guevara.)
RA1	Expression – the physically realized.

RA/PMT’s central thesis: QM, GR, emergent spacetime and cosmology all possess RA4 and RA2 but *no* RA1.5. The absence of this selector is claimed to be the common root of the measurement problem, the initial-conditions problem, fine-tuning, the arrow of time, and the cosmological constant. RA/PMT “completes” the theories by adding RA1.5 – without altering their equations.

.2 Direction A: FFGFT evaluates RA/PMT (six-axis operator)

We apply the comparison operator from Doc. 265: two frameworks are compared along six axes (shared invariants, operators, mathematical structures, predictions, physical mechanisms, hardware implications). On each axis: shared, partial, or not.

Finding, Direction A: On four of six axes there is *no* contact, on two only contact at the meta-level (axes 2 and 3). By the classification of Doc. 265 this is not even Level 3 (complementary in the usual sense) – it is **category difference**: the two frameworks do not answer the same kind of question. The only genuine overlap is methodological: both separate cleanly what follows rigorously from what remains open. FFGFT does this through its three-layer methodology (cores proved from ξ > bridges proved algebraically > reductions as plausibility), RA/PMT through RA4/RA3/RA2/RA1.5/RA1. This discipline is congenial and related – but it is a *stance*, not shared physics.

.3 Direction B: the RA/PMT grid applied to FFGFT

Guevara applies his grid to other theories himself (QFT, GR, emergent spacetime, IIT, ...). The logical consequence: one can apply it to FFGFT too. This is illuminating.

Table 1: FFGFT $\leftrightarrow$ RA/PMT along the six comparison axes. $\boxtimes$ = shared, $\sim$ = only at the meta-level, $\times$ = not shared.

Comparison axis	Status	Note
1. Invariants	$\times$	RA/PMT supplies no invariant; FFGFT: $\tilde{T} \cdot m = 1$, winding number.
2. Operators	$\sim$	Structural only: RA/PMT's "selector" vs. FFGFT's recursion $\mathcal{R}$ (Doc. 203) – but $\mathcal{R}$ is defined, RA1.5 is not.
3. Mathematical structure	$\sim$	Shared only as <i>layering discipline</i> : FFGFT's cores $>$ bridges $>$ reductions $\approx$ RA/PMT's RA4 $>$ RA2 $>$ RA1.
4. Predictions (quantitative)	$\times$	RA/PMT produces not a single number; FFGFT: peak ratios, H_0 , masses from ξ .
5. Physical mechanisms	$\times$	RA1.5/CSM is pre-physical and unspecified; FFGFT mechanisms are geometrically concrete.
6. Hardware implications	$\times$	RA/PMT makes no statement about devices/measurements.

Layer	FFGFT assignment
RA4	The T^4 mode space: winding numbers $\vec{n} \in \mathbb{Z}^4$, the possible modes.
RA3	Open points, already explicitly declared in FFGFT (P-register, Doc. 190): e.g. the exponent $41/4$ (R20), the rigorous C_t projection (R30).
RA2	The derivations from ξ : Jacobi degeneracy, Bose-Einstein, 3D projection, recursion operator $\mathcal{R}$ (Doc. 203).
RA1.5	Here RA/PMT asks: where is your selector?
RA1	Observable: CMB peak ratios, H_0 , mass relations.

The decisive point – FFGFT has no RA1.5, and needs none. Guevara’s grid diagnoses a “missing selector” in every theory. For FFGFT the answer is not “the selector is missing” but “*the premise does not hold*”:

- RA1.5 presupposes that there is a space of *many possible* universes/configurations (RA4) from which *one* is selected. That is the multiverse-like assumption RA/PMT carries tacitly (as Λ CDM carries its initial conditions).
- FFGFT is **static**: no Big Bang, no $t = 0$, no initial state to be selected. The fractal T^4 torus is not *one selected* among many configurations – it is *the* structure. There is no blueprint space for a selector to choose from.
- Where Guevara’s cosmology (Part III) reads the Big Bang, inflation and the low initial entropy as “the expression of a selected blueprint”, FFGFT says: these events are not initial conditions to be selected, because there is no beginning (Doc. 267, inverted initial condition).

What Direction B exposes: The RA/PMT grid is not neutral. It carries a hidden assumption – that *every* theory needs a selector between possibility and actuality. FFGFT is precisely the counterexample: a theory that deliberately lacks this layer because it does not make the possibility/actuality split. The “deficiency” the grid diagnoses is, from the FFGFT side, not a deficiency but the absence of a superfluous assumption.

This is methodologically the most valuable finding of the whole comparison: it shows that RA1.5 itself – by Guevara’s own RA3 definition – is a *residual*. “CSM selects the low-entropy configuration” renames the puzzle instead of solving it: there is no equation, no selection criterion, no prediction. A named selector without operationalization meets exactly the RA3 condition (“lacks a defined mechanism”) that RA/PMT charges against other theories.

.4 Synthesis: two opposite poles of the same question

Taken together, the two directions give a clear picture. FFGFT and RA/PMT are neither competitors nor variants – they are **opposite poles of the same basic question**: how does the observed reality come about?

	RA/PMT	FFGFT
Basic assumption	many possible universes (RA4), one is selected	one static fractal T^4 , no selection
Selector RA1.5	necessary but undefined (CSM/PSM/HPM)	absent, not needed
Big Bang, inflation, initial entropy	real events whose configuration is selected	do not exist ($t = 0$ drops out)
Λ , fine-tuning	selected parameters	premise rejected (no dark sector; redshift = fractal path-lengthening, Doc. 190 R14)
Delivers numbers?	no (meta-scheme)	yes ($\xi \rightarrow$ peaks, H_0 , masses)
Strength	layering discipline, conceptual hygiene	parameter-free, falsifiable predictions

Where RA/PMT posits a missing layer, FFGFT says: the layer is not missing, it is not needed – the premise (many possible configurations, one selected) is itself a Λ CDM-like assumption. This is the same methodological line as the “inverted initial condition” and the symmetric circularity of Doc. 267: RA/PMT’s CSM is essentially Λ CDM’s initial conditions with a metaphysical selector in front; FFGFT removes the need for both.

Fair appraisal. This is not meant to disparage RA/PMT. As a *meta-scheme* – as a discipline for sorting a theory’s terms cleanly into blueprint, residual, mapping, mechanism and expression – it is a useful diagnostic tool, and its honesty in declaring residuals deserves credit. It shares with FFGFT the stance “inspectable rather than persuasive” (Doc. 265). The disagreement is not in the method but in the premise: whether reality is a selection from the possible (RA/PMT) or a single, static, fractal structure with no selection (FFGFT).

Residuals of this comparison (honestly declared).

- The comparison treats RA/PMT as a meta-scheme. Should Guevara later operationalize RA1.5 (give an equation, a selection criterion), Direction A would need re-evaluation – there might then be axis-4 contact.
- Direction B assumes FFGFT’s static premise is correct. That is an FFGFT-internal position, not a proven theorem; an RA/PMT proponent would read it as itself a selected configuration. Both readings remain side by side (cf. the provenance principle, Doc. 265).

Related: Doc. 265 (comparison operator, six axes), Doc. 267 (cosmological degeneracy, inverted initial condition), Doc. 203 (recursion operator $\mathcal{R}$), Doc. 190 (R14, R20, R30). RA/PMT: J. T. Guevara, *Architectural Completion* (Parts I–III), June 2026.